

Q Explain Regular Expression & regular language. (3 Marks)

Regular Expression: specific type of expression used to represent language

It is an algebraic notation used to describe regular language.

* Symbols used in RE :

$a \rightarrow$ single symbol

$+$ $\rightarrow$ union OR

$ab \rightarrow$ concatenation

$*$ $\rightarrow$ Kleene star (zero or more repetitions)

$\epsilon \rightarrow$ empty string

* Examples of RE.

(1) $(0+1)^*$

Meaning : all possible strings of 0 and 1

strings generated : $\epsilon, 0, 1, 00, 01, 10, 11, 101$

(2) $a(b+c)$

Meaning : strings starting with 'a' followed by either 'b' or 'c'

strings generated : ab, ac

Regular language: set of strings that can be described by a RE or accepted by a finite automata.

ex. $L = \{\text{all strings that end with 'a'}\}$

$\therefore L = \{a, ba, aa, bba, aba, \dots\}$

for the given language L , language L is regular

because RE exists for the given language L .

$\rightarrow$ R.E. = $(a+b)^*a$

ex $L_1 = \{a^n \mid n \geq 0\}$

$L_1 = \{\epsilon, a, aa, aaa, \dots\}$

RE = a^*

so, given language L_1 is regular

Q.

Given two languages L_1 and L_2 defined over $\Sigma = \{a, b\}^*$, L_1 accepts palindrome strings and L_2 accepts strings with equal number of 0's and 1's.

Which one of these languages is regular? Give reasons. (3 Marks)

L_1 = set of palindrome strings

L_2 = set of strings with equal number of 0's and 1's.

① Language L_1 (Palindromes)

A palindrome reads the same forward and backward.

example: aba, bb, 0110, ...

To check whether a string is palindrome, a machine must remember first half of the string and compare it with the second half.

Finite automata have limited memory, so this is not possible. So, language L_1 is not regular language because FA is not possible for it.

→ using Pumping Lemma also palindrome language can be proved non-regular.

Hence, L_1 is not regular.

② Language L_2 (equal no of 0's and 1's)

ex 01, 0011, 1010, ...

→ To accept this language, the machine must count the number of 0's and 1's and compare them.

A finite Automata cannot store unlimited counts.

→ using Pumping Lemma, this language L_2 can be proved non-regular.

Hence, L_2 is not regular.

Q. show that language 'pal' of palindrome is not regular. (3 Marks)

⇒ Let $PAL = \{w \mid w = w^R\}$
 w^R is reverse of string w .

Proof using Pumping Lemma:

Assume that PAL is regular.

By pumping lemma, there exists a pumping length 'p'

choose a string: $S = 0^p 1 0^p$

this string is a palindrome and belongs to PAL.

Now divide: $S = xyz$

$$|xy| \leq p$$

$$|y| > 0$$

$$xy^iz \in L \text{ for all } i \geq 0$$

$$\Rightarrow S = 0^p 1 0^p$$

$$\text{if } p=3: 0001000$$

$$|xy| \leq p \quad |y| > 0$$

$$|000| \leq p \quad |00| > 0 \rightarrow 2 > 0$$

$$3 \leq 3 \quad \therefore x=0, y=00, z=1000$$

$$\Rightarrow \text{Pump } y$$

$$xy^iz \in L$$

$$i=0: xy^0z = xz \\ = 01000$$

Original: 0001000 $\rightarrow$ reverse is 0001000

after pump: 01000 $\rightarrow$ reverse is 00010 $\rightarrow$ different.

⇒ Contradiction

Pumping lemma said: after pumping/removing y , string should still remain in language.

But after removing: $xz \notin PAL$

$\therefore$ Our assumption is wrong.

Hence, PAL is not regular.

Q. Explain the importance of distinguishable strings and equivalent classes' w.r.t. regular languages.
(7 marks)

⇒ **Distinguishable strings** : Two strings are distinguishable if there exists some suffix that makes one accepted and the other rejected.

ex. $L = \{\text{strings ending with } 1\}$

Accepted language strings : 1, 01, 101
Rejected language strings : 0, 10, 100

} $\begin{matrix} 1 \& 0, \\ 01 \& 10, \\ 101 \& 10 \end{matrix}$ are distinguishable strings.

→ string from accepted : $x = 1$

rejected : $y = 0$

suppose we add suffix '0' at each string,

$\begin{matrix} x' = 10 \\ y' = 00 \end{matrix}$ } both are rejected by given language.

so, for the string x & y , both are not distinguishable strings.

ex $L = \{\text{strings ending with } ab\}$

accepted strings = $\{ab, aab, bab, bbab, \dots\}$

→ take two strings : Not necessarily from given language.

$x = a \rightarrow$ rejected by L

$y = b \rightarrow$ rejected by L

add suffix b at both : $x = ab \rightarrow$ accepted by L

$y = bb \rightarrow$ rejected by L

∴ string 'a' & 'b' are distinguishable strings.

⇒ Equivalent classes: Group of strings that behave the same way.

ex $L = \{ \text{strings ending with } 1 \}$

Accepted strings = $\{ 1, 01, 101, 111 \}$

Rejected strings = $\{ 0, 10, 110, 000, 010 \}$

→ if we add same suffix to all accepted strings:
suppose add 0 to each: $\{ 10, 010, 1010, 1110, \dots \}$
↳ all will be rejected by L .

So, all strings after adding suffix belong to same group: Equivalent class.

→ Take two strings from rejected strings:

$x = 10, y = 000$

add suffix 1 in each.

$\therefore x' = 101 \rightarrow \text{accepted by } L$
 $y' = 0001 \rightarrow \text{accepted by } L$ } same behavior

add suffix 0 in each.

$x' = 100 \rightarrow \text{rejected by } L$
 $y' = 0000 \rightarrow \text{rejected by } L$ } same behavior

$\therefore$ Equivalent class.

* Importance: Distinguishable strings and equivalence classes
helps in DFA minimization

- used to remove unnecessary states in DFA.
- helps to identify non-regular languages.
- helps to classify strings with similar behavior.